Safety Data Sheet

Creation Date: November, 1985
Revision Date July, 2014

1. Product and Company Identification

Product Code: A 190
Product Name: SphereOX®
Chemical Family: Metal Oxide
Synonym(s): SphereOX® Fines, SphereOX Coarse®
C.A.S. Number: 7439-89-6 & 1309-37-1 & 1317-61-9
Formula: Fe_3O_4

Manufacturer Information:
Chesapeake Specialty Products, 5055 North Point Boulevard, Baltimore, MD 21219
Business Telephone: (410) 388-5055

For Additional Information, contact:
Occupational Health & Safety Division (410) 388-5055

2. Composition/Information on Ingredients

Non-Hazardous Ingredients:

Components:	C.A.S.	%	Exposure Limits (8 Hrs. TWA) OSHA PEL	ACGIH TLV
Fe, Fe_2O_3, Fe_3O_4	7439-89-6 & 1309-37-1 & 1317-61-9	>99%	Not Established	Not Established

3. Hazards Identification

**********Emergency Overview**********

Dry, Black powder with little to no odor. Will not burn or react. Long-term inhalation can cause lung irritation or siderosis. Packaging material can burn or melt in fire, producing toxic smoke and fumes.

****

HMIS Designation: **Health**: 1 **Flammability**: 0 **Reactivity**: 0 **Personal Protection**: 0

Potential Health Effects:

Eye Contact: Non-irritating to the eyes. Excessive exposure to airborne nuisance dust may reduce visibility and/or cause unpleasant deposits.

Skin Contact: Will not irritate skin and is not likely to cause allergic skin reaction. Irritation to skin or mucous membranes can occur by direct mechanical action.

Ingestion: Small amounts (a tablespoon) swallowed

Inhalation: As with all dusty materials, inhalation may cause respiratory irritation, sneezing, coughing, and runny nose. Wear respirator and avoid breathing dust.

Human Effects and symptoms of overexposure:

Acute: To date, adverse health effects from exposure have not been reported among workers using this oxide. On the basis of Animal Toxicity data (see Section 11), we would expect this product to be non-irritating to the eyes and skin and essentially non-toxic by ingestion. However, excessive air born nuisance dust may reduce visibility and/or cause unpleasant deposits in the eyes, ears and nose.

Chronic: None Known

Other Effects: None Known

Carcinogenicity: IARC: Not Listed NTP: Not Listed OSHA: Not Regulated

4. First Aid Measures

Eye Contact: Flush eyes with water, lifting eyelids periodically. Remove contact lenses and discard. Continue flushing for 15 minutes or until eyes return to normal. Get medical attention if irritation develops or persists.

Skin Contact: Wash with soap and water. Get medical attention if irritation develops or persists. Wash clothing before re-use.

Ingestion: Swallowing less than one ounce (30g) will not cause harm. For larger amounts, do not induce vomiting, instead give one or two glasses of water (8-10 oz) to drink. Contact medical personnel or poison control center immediately.

Inhalation: Move from dusty area to fresh air. Get medical attention for any difficulty breathing. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Get immediate medical attention.

5. Firefighting Measures

Flammable Properties:	Not flammable
Flash Point:	Will not flash
Upper Explosive Limit (UEL):	Will not explode
Lower Explosive Limit (LEL):	Will not explode
Auto-Ignition Temperature:	Will not auto-ignite.
Extinguishing Media:	This product is not combustible or flammable. Use extinguishing agents that are suitable to the surrounding fire; water spray, dry chemical, foam or CO_2.
Fire Fighting Instructions:	Firefighters should be equipped with self-contained breathing apparatus to protect against potentially toxic and/or irritating fumes and smoke inhalation.

6. Accidental Release Measures

Small Spill: If dust is generated, use appropriate respiratory protection. Vacuum or scoop material into an appropriately marked container for re-use or disposal. Avoid excessive generation of dust.

Large Spill: Use recommended protective clothing and respiratory protection. Use shovel to reclaim material. Vacuum or scoop material into an appropriately marked container for re-use or disposal. Avoid excessive generation of dust. It is more effective to clean this product while dry by vacuuming or sweeping. However, spill area can be washed with water. Collect the wash water for approved disposal. Prevent runoff from entering storm sewers and ditches which lead to natural waterways.

7. Handling and Storage

Storage: Store at ambient temperatures away from food and beverages, excessive heat or flame sources (furnace, kilns, boilers etc.). Keep separate from substances subject to catalytic decomposition by dust e.g. peroxides.

Handling: Avoid breathing dust. Void getting in eyes or on skin. Wash hands thoroughly after handling. Avoid contact with moisture. Re-seal bag immediately after use. Pallets may be wrapped in polyethylene plastic. Removal may cause electrostatic spark; therefore removal of the wrap should not be done in the presence of flammable vapors.

Storage Temperature (Min/Max)	Ambient/50°C (122°F)
Shelf Life	Unlimited in closed container
Special sensitivity	None
Other Precautions	None

8. Exposure Controls/Personal Protection

Engineering Controls: Maintain air levels below the recommended exposure limit using process enclosure and exhaust ventilation if necessary. Supply sufficient replacement air to make up for air removed exhaust systems. If engineering controls and work practices are not effective in controlling exposures, appropriate personal protective equipment including a NIOSH/OSHA approved dust respirator should be worn.

Eyes: Wear safety glasses with side shields or goggles. Eye wash stations should be available in workplace.

Skin: Wear body-covering clothing closed at wrists and ankles. Rubber, PVC or Leather gloves are suggested to facilitate personal hygiene.

Respiration Protection: Workplace ambient dust concentrations should be monitored and if the recommended exposure limit is exceeded, a NIOSHMSHA approved respirator with dust filter should be worn.

Other: Emergency eye wash stations should be available. Educate and train employees in safe use and handling of any chemicals on premise.

Work/Hygiene Practices: Employees should wash their hands and face before eating, drinking or using tobacco products.

9. Physical and Chemical Properties

Appearance	Solid Black Powder
Odor	Odorless
Physical State	Dry Powder
pH	7.0 in H_2O in aqueous suspension.

Vapor Pressure	Not a vapor
Vapor Density	Not a vapor
Boiling Point:	Not Applicable
Freezing Point:	Not Applicable
Melting Point:	Greater than 1000° C (1832°F)
Solubility in Water:	Insoluble
Specific Gravity (g/ml):	5.05 @ 20°C
Bulk Density (kg/m^3):	2900 @ 20°C
Particle Size (microns):	64-425
Volatile Organic Compounds (VOC):	None
Chemical Formula:	Fe_3O_4

10. Stability and Reactivity

Chemical Stability:	Stable Keep away from flames and heat. Exposure to excessive heat can cause containment to break down or burn away.
Incompatibilities:	Mineral acids
Hazardous Decomposition Products:	None
Hazardous Polymerization:	Will not occur

11. Toxicological Information

Eyes:	Not irritating to rabbit eyes.
Skin:	Not irritating to rabbit dermal, LD50 not established for this product
Ingestion:	Non irritating. The oral, LD50 for rats is >5000 mg/l
Inhalation:	Non Irritating. LC50 is not established for product.
Subchronic:	Data not established
Chronic/Carcinogenicity:	Data not established.
Other (Mutagenic, Teragenic, Reproductive tests)	None known

12. Ecological Information

Ecotoxilogical Information:	Fish Toxicity: Lco >1000mg/l
Chemical Fate Information:	No appreciable bioconcentration is expected in the environment

Disclaimer

This form is being provided solely as general information and should not be construed as a determination that the product is a hazardous chemical. All sales of this product are subject to CHESAPEAKE'S Standard Terms and Conditions of Sale.

CHESAPEAKE MAKES NO WARRANTIES, EXPRESS OR IMPLIED, INCLUDING THE IMPLIED WARRANTY OF MERCHANTABILITY, ANY IMPLIED WARRANTY OF FITNESS FOR A PARTICULAR PURPOSE OR ANY IMPLIED WARRANTIES OTHERWISE ARISING FROM COURSE OF DEALING OR TRADE.